PN junction, P/N type, depletion

1. Q: What happens if P and N regions are not doped properly?
   A: Depletion width and barrier potential become unsuitable, causing poor switching and wrong breakdown voltage.[1]

2. Q: What if P and N doping are made equal in a power diode?
   A: Depletion region spreads symmetrically, but blocking capability and carrier storage become less optimized.[1]

3. Q: What if N side is more lightly doped than P side in a rectifier?
   A: Depletion region extends more into N, giving higher reverse blocking voltage.[1]

4. Q: Why is N side usually lightly doped for high- voltage devices?
   A: A lightly doped region tolerates higher electric field before breakdown.[1]

5. Q: What happens if we use P- type instead of N- type for the drift region?
   A: Majority carriers change to holes, mobility reduces, so on- state loss increases.[1]

6. Q: Why is N- type preferred for high- speed conduction paths?
   A: Electrons have higher mobility than holes, giving lower resistance and faster switching.[1]

7. Q: What happens if the depletion layer does not widen in reverse bias?
   A: Reverse voltage cannot be blocked and breakdown occurs at low voltage.[1]

8. Q: Why is depletion region called **insulating**?
   A: It has almost no free carriers, so it behaves like a dielectric.[1]

9. Q: What happens if barrier potential is very low in a diode?
   A: Diode turns ON at very small forward voltage, increasing leakage and power loss.[1]

10. Q: Why does silicon have about 0.7 V barrier while germanium has ~0.3 V?
    A: Because of different bandgaps and intrinsic carrier concentrations.[1]

11. Q: What if we increase temperature of a PN junction?

    A: Reverse saturation current rises and breakdown occurs at lower voltage.[1]

12. Q: Why does depletion width increase with reverse voltage?

    A: Reverse bias pulls carriers away from the junction, extending the space- charge region.[1]

13. Q: What happens if we suddenly remove reverse bias?

    A: Depletion shrinks and junction returns to near- equilibrium width.[1]

14. Q: Why is depletion region critical for switching speed?

    A: Its growth and shrink define how fast the junction can transition between OFF and ON.[1]

15. Q: What happens if minority carrier lifetime is very high in PN junctions?

    A: Stored charge increases, causing slow turn- off.[1]

Rectifier diode basics, biasing, polarity

16. Q: What happens if we forward bias a rectifier diode?

    A: Depletion narrows, barrier potential reduces, and large current flows.[1]

17. Q: What happens if we reverse bias the rectifier diode?

    A: Depletion widens and only tiny leakage current flows.[1]

18. Q: Why does a rectifier diode behave like a switch?

    A: It approximates closed circuit in forward bias and open circuit in reverse bias.[1]

19. Q: What happens if we reverse the polarity of a half- wave rectifier input?

    A: Conduction shifts to the opposite half cycle or no output if orientation is wrong.[1]

20. Q: What happens if we use a diode backwards in a DC supply?

    A: It blocks current, so load receives no DC.[1]

21. Q: Why can a silicon rectifier dissipate power even when ON?

    A: Because of around 0.7 V drop across it at high current.[1]

22. Q: What happens if diode rating is lower than the circuit current?
    A: Junction overheats and may fail short or open.[1]

23. Q: Why is a standard rectifier not used at very high frequency?
    A: Its reverse recovery is slow, causing large switching loss.[1]

24. Q: What happens if reverse voltage exceeds the diode's breakdown rating?
    A: Avalanche breakdown and possible permanent damage.[1]

25. Q: Why is leakage current tolerated in reverse bias?
    A: It is very small and usually does not affect load or regulation much.[1]

26. Q: What if we connect two rectifier diodes in series for reverse blocking?
    A: Effective breakdown voltage increases (with balancing measures).[1]

27. Q: Why are rectifier diodes often silicon rather than germanium?
    A: Better temperature stability and higher reverse voltage capability.[1]

28. Q: What happens if we remove the load in a rectifier circuit?
    A: Output voltage rises toward peak input minus diode drops.[1]

29. Q: Why is ripple large in a half‑ wave rectifier?
    A: Because current flows only during one half cycle.[1]

30. Q: What happens if we add a filter capacitor after the rectifier?
    A: Ripple reduces and output becomes smoother DC.[1]

Advanced diodes: fast, Schottky, PIN

31. Q: Why use fast‑ recovery diodes in SMPS?
    A: To reduce reverse recovery time and switching losses at high frequency.[1]

32. Q: What happens if a standard diode is used where ultrafast is needed?
    A: Excess reverse recovery current and heating, possibly failure.[1]

33. Q: Why does a Schottky diode have low forward drop?

    A: It is a metal–semiconductor junction with majority-carrier conduction.[1]

34. Q: What happens if we replace a PN diode with a Schottky in a low-voltage supply?

    A: Efficiency improves due to lower forward voltage.[1]

35. Q: Why Schottky is not used for very high blocking voltages?

    A: Its reverse leakage grows rapidly with voltage.[1]

36. Q: What happens if we use a Schottky where low leakage is critical?

    A: Excess leakage can cause standby losses and wrong biasing.[1]

37. Q: Why is a PIN diode used for high-voltage blocking?

    A: Its intrinsic layer supports wide depletion and high breakdown voltage.[1]

38. Q: What happens if we remove the intrinsic layer in a PIN diode structure?

    A: Blocking capability drops and it behaves more like a normal PN diode.[1]

39. Q: Why fast diodes have shorter carrier lifetime?

    A: Recombination centers are introduced to clear stored charge quickly.[1]

40. Q: What happens if carrier lifetime is too short?

    A: Reverse recovery improves but forward voltage and on-state loss increase.[1]

Half-wave and full-wave rectifiers

41. Q: What if we interchange diode connection in a half-wave rectifier?

    A: It will rectify the opposite half cycle.[1]

42. Q: Why full-wave rectifier gives higher efficiency?

    A: It uses both half cycles of AC input.[1]

43. Q: What happens if one diode in a bridge rectifier is open?

    A: Output becomes distorted or drops to half-wave behavior.[1]

44. Q: Why bridge rectifier has two diode drops in series?

    A: Current path always includes two diodes between AC and DC terminals.[1]

45. Q: What happens if transformer secondary is center- tapped but used with bridge?

    A: Copper is wasted and secondary utilization is poor.[1]

46. Q: What if we reverse the load polarity in a bridge rectifier?

    A: DC polarity across the load reverses.[1]

47. Q: Why full- wave output frequency is double input frequency?

    A: Each half cycle (positive and negative) produces a positive output pulse.[1]

48. Q: What happens if filter capacitor value is too large?

    A: Better ripple but high inrush current and diode stress at charging peaks.[1]

49. Q: Why high- current rectifiers are mounted on heat sinks?

    A: To remove heat due to conduction losses and prevent thermal runaway.[1]

50. Q: What happens if thermal resistance is high?

    A: Junction temperature rises quickly, reducing lifetime.[1]

BJT structure: replacing P/N, junction conditions

51. Q: What happens if the base of a BJT is thick?

    A: More recombination, lower gain, and poorer switching.[1]

52. Q: Why is the BJT base lightly doped and thin?

    A: To let most injected carriers reach the collector.[1]

53. Q: What happens if emitter is not heavily doped?

    A: Injection efficiency falls and current gain reduces.[1]

54. Q: Why is collector moderately doped and large area in power BJT?

    A: To handle high voltage and dissipate heat.[1]

55. Q: What happens if both EB and BC junctions are reverse biased?

    A: BJT goes to cut- off and acts like open switch.[1]

56. Q: What if EB is forward biased and BC is reverse biased?

    A: BJT operates in active region (amplifier).[1]

57. Q: What if both EB and BC junctions are forward biased?

    A: BJT goes into saturation (fully ON switch).[1]

58. Q: Why is saturation used in power switching?

    A: It gives low VCEsat and low conduction loss.[1]

59. Q: What happens if base current is insufficient in a switch application?

    A: Transistor enters quasi- saturation, causing high VCE and heat.[1]

60. Q: Why NPN is preferred over PNP for high- power BJTs?

    A: Electrons as majority carriers give better speed and conduction.[1]

61. Q: What happens if we swap collector and emitter connections?

    A: Device works poorly with low gain and low breakdown voltage.[1]

62. Q: Why does BJT need continuous base drive in ON state?

    A: Collector current is proportional to base current.[1]

63. Q: What happens if base current is abruptly removed?

    A: Stored charge must recombine, producing slow turn- off.[1]

64. Q: Why is stored charge a drawback in power BJTs?

    A: It slows turn- off and increases switching loss.[1]

65. Q: What happens if we speed up recombination in BJT?

    A: Switching improves but gain and conduction efficiency may worsen.[1]

66. Q: Why are BJTs called **bipolar**?

    A: Both electrons and holes participate in conduction.[1]

BJT as a switch: polarity, modes

67. Q: What happens if base‑emitter is reverse biased in a BJT switch?
    A: Transistor remains OFF (cut‑off).[1]

68. Q: Why is VBE around 0.7 V needed for silicon BJTs?
    A: This forward bias overcomes the PN barrier potential.[1]

69. Q: What if VBE is below about 0.7 V in a power BJT?
    A: Transistor remains OFF or only slightly conducts.[1]

70. Q: What happens to VCE in saturation?
    A: VCE drops to a small value (~0.2 V).[1]

71. Q: Why does BJT behave like closed switch in saturation?
    A: Because VCE is low and collector‑emitter resistance is small.[1]

72. Q: What if base current is higher than required for saturation?
    A: Device is in deep saturation; more drive but no major VCE reduction.[1]

73. Q: Why is deep saturation sometimes avoided?
    A: It increases stored charge and slows turn‑off.[1]

74. Q: What happens if collector voltage becomes negative in NPN switch?
    A: Device may go into reverse‑biased breakdown or damaged region.[1]

75. Q: Why is BJT slower than MOSFET in switching?
    A: Because of minority carrier storage in the base.[1]

76. Q: What happens if we try to switch BJT at very high frequency?
    A: Switching losses dominate and device overheats.[1]

MOSFET basics: gate, channel, P/N swap

77. Q: What happens if VGS is below threshold in enhancement MOSFET?

    A: No channel forms and device stays OFF.[1]

78. Q: Why is MOSFET called **voltage‑controlled**?

    A: Gate current is negligible; only gate‑source voltage controls conduction.[1]

79. Q: What happens if VGS exceeds threshold?

    A: An inversion channel forms and drain current flows.[1]

80. Q: Why is gate isolated by SiO_2?

    A: To form a capacitor and prevent DC current into gate.[1]

81. Q: What happens if oxide is damaged?

    A: Gate leaks, device may fail or lose control characteristics.[1]

82. Q: What if we swap N‑channel MOSFET with P‑channel in the same low‑side circuit?

    A: Polarities invert; device may never turn ON correctly.[1]

83. Q: Why N‑channel MOSFET has lower RDS(on) than P‑channel?

    A: Electron mobility is higher than hole mobility.[1]

84. Q: What happens if we use P‑channel MOSFET for high‑current low‑side switching?

    A: Higher conduction losses due to higher RDS(on).[1]

85. Q: Why MOSFET is called **unipolar**?

    A: Conduction depends mainly on one type of carrier (electrons or holes).[1]

86. Q: What happens if gate voltage is left floating?

    A: Device can randomly turn ON/OFF due to noise and capacitance.[1]

87. Q: Why must gate‑source voltage be limited?

    A: Exceeding maximum VGS can puncture gate oxide.[1]

88. Q: What happens if source and drain of MOSFET are interchanged physically?
    A: Body diode orientation may be wrong and RDS(on) characteristics degrade.[1]

89. Q: Why is there an inherent body diode in power MOSFETs?
    A: Because of the PN junction between P- body and N- drain.[1]

90. Q: What happens if body diode conducts in an H- bridge unexpectedly?
    A: Unwanted current paths and shoot- through can occur.[1]

Enhancement vs depletion MOSFETs

91. Q: What happens at VGS = 0 in an N- channel enhancement MOSFET?
    A: Device is normally OFF.[1]

92. Q: What happens at VGS = 0 in an N- channel depletion MOSFET?
    A: Device is normally ON.[1]

93. Q: Why enhancement MOSFETs are more common in power circuits?
    A: Normally OFF behavior is safer for fail- safe designs.[1]

94. Q: What happens if VGS is made negative for N- channel depletion MOSFET?
    A: Channel is depleted and current reduces or stops.[1]

95. Q: Why depletion MOSFETs are rare in SMPS?
    A: Their normally ON nature complicates startup and protection.[1]

96. Q: What happens if we drive an enhancement MOSFET with VGS slightly above threshold only?
    A: It works in high- resistance region, causing large losses.[1]

97. Q: Why is high gate voltage (10–12 V) used for power MOSFETs?
    A: To fully enhance the channel and minimize RDS(on).[1]

98. Q: What happens if we use logic- level gate drive on a non- logic MOSFET?

A: Device partially turns ON with high conduction loss.[1]

99. Q: Why MOSFET has ohmic and saturation regions?

A: Channel behaves like variable resistor at low VDS and like current source at higher VDS.[1]

100. Q: What happens if MOSFET is operated in linear region for long time?

A: Large dissipation and risk of thermal failure.[1]

MOSFET switching: polarity, dynamic

101. Q: What happens when VGS rises from 0 to above threshold?

A: Gate capacitance charges, channel forms, current rises and VDS falls.[1]

102. Q: Why is gate charge more important than just input capacitance?

A: Total gate charge determines actual time to switch.[1]

103. Q: What happens if gate driver is weak (high source resistance)?

A: Slow edges, higher switching losses, and EMI problems.[1]

104. Q: Why MOSFET turns OFF fast compared to BJT?

A: It does not store minority carriers, only capacitive charge.[1]

105. Q: What happens if we suddenly short gate to source?

A: Gate charge quickly discharges and MOSFET turns OFF.[1]

106. Q: Why is there a Miller plateau in VGS waveform?

A: Gate voltage stays almost constant while VDS changes due to CGD charge.[1]

107. Q: What if drain voltage is very high while turning OFF?

A: dv/dt across CGD can induce spurious gate voltage and false turn- on.[1]

108. Q: Why gate resistors are used?

A: To control switching speed and limit dv/dt and ringing.[1]

109. Q: What happens if gate resistor is zero?

A: Very fast transition, high EMI and possible overshoot.[1]

110. Q: Why MOSFET is preferred over BJT in high- frequency SMPS?

A: Much faster switching and lower switching loss.[1]

IGBT structure and P/N role

111. Q: What happens if the P+ collector layer in IGBT is removed?

A: Device loses bipolar behavior and becomes more MOSFET- like.[1]

112. Q: Why IGBT has N- drift region?

A: To support high blocking voltage.[1]

113. Q: What happens if drift region is too thin?

A: Blocking voltage reduces and breakdown occurs early.[1]

114. Q: What if drift region is too thick?

A: On- state voltage increases due to higher resistance.[1]

115. Q: Why IGBT's emitter is N- type?

A: To form MOS channel with P- body for electron injection.[1]

116. Q: What happens if P- body is not properly doped?

A: Threshold voltage and blocking capability become unpredictable.[1]

117. Q: Why IGBT is "MOSFET at input, BJT at output"?

A: Gate controls MOS channel, which drives a bipolar conduction path.[1]

118. Q: What happens if there were no P+ collector in IGBT?

A: No hole injection and no conductivity modulation, so higher losses.[1]

119. Q: Why IGBT can handle high voltage with low conduction loss?

A: N- drift gives voltage blocking while conductivity modulation lowers on- resistance.[1]

120. Q: What happens if minority carrier lifetime is very high in IGBT?

A: Tail current during turn‑ off increases and switching loss rises.[1]

IGBT operation, polarity, switching

121. Q: What happens if VGE < threshold in IGBT?

A: MOS channel does not form and device remains OFF.[1]

122. Q: What happens if VGE is raised to about 10–15 V?

A: Channel fully forms and device turns ON with low VCEsat.[1]

123. Q: Why IGBT is voltage‑ controlled?

A: Gate draws negligible steady‑ state current; only voltage controls channel.[1]

124. Q: What happens if collector‑ emitter voltage exceeds rating in OFF state?

A: Avalanche breakdown can destroy the device.[1]

125. Q: Why IGBT tail current appears during turn‑ off?

A: Stored minority carriers in drift region recombine slowly.[1]

126. Q: What happens if switching frequency is too high for an IGBT?

A: Tail current causes large switching losses and excess heating.[1]

127. Q: Why IGBT is preferred over MOSFET at high voltage and moderate frequency?

A: Lower conduction loss than MOSFET at high voltage.[1]

128. Q: What happens if IGBT is used in very high‑ frequency SMPS?

A: Tail current makes it inefficient compared to MOSFET.[1]

129. Q: Why IGBT needs careful gate turn‑ off control?

A: To manage tail current and avoid overvoltage.[1]

130. Q: What happens if gate is driven negative at turn‑ off?

A: Faster removal of charge and reduced tail, within limits.[1]

Thyristor/SCR, TRIAC, GTO logic

131. Q: What happens if gate of SCR is never triggered?

A: Device stays OFF unless forward voltage exceeds breakover.[1]

132. Q: What happens once SCR is triggered ON?

A: It latches ON and continues conducting until current falls below holding current.[1]

133. Q: Why SCR is called semi‑ controlled?

A: Gate controls turn‑ on but not turn‑ off.[1]

134. Q: What if we reverse the polarity across SCR?

A: Reverse‑ biased; only small leakage flows until reverse breakdown.[1]

135. Q: Why SCR is not ideal for forced commutation at high power?

A: Turn‑ off requires current to drop below holding value by external circuits.[1]

136. Q: What happens if gate current is too small in SCR?

A: It may fail to trigger or trigger at higher forward voltage.[1]

137. Q: Why TRIAC can conduct in both directions?

A: It is effectively two inverse parallel SCRs integrated.[1]

138. Q: What happens if TRIAC is used with highly inductive load?

A: Commutation becomes difficult and mis‑ triggering can occur.[1]

139. Q: Why TRIAC is used for AC lamp dimmers?

A: It can control conduction angle for both half cycles.[1]

140. Q: What happens if GTO gate is not given negative pulse to turn OFF?

A: It may not turn OFF reliably and remains conducting.[1]

141. Q: Why GTO is called fully controlled?

A: Gate controls both turn‑ on and turn‑ off.[1]

142. Q: What happens if gate drive of GTO is weak at turn‑ off?

A: It may fail to commutate and overheat.[1]

Classification & choice of devices

143. Q: Why diodes are called two‑ terminal devices?

A: They only have anode and cathode.[1]

144. Q: Why BJT, MOSFET, IGBT are three‑ terminal?

A: They have separate control terminal besides two power terminals.[1]

145. Q: What happens if gate/base is shorted to emitter/source?

A: Device is forced OFF (for normal polarities).[1]

146. Q: Why power devices are also classified as fully controlled or semi‑ controlled?

A: Based on whether both turn‑ on and turn‑ off are controllable.[1]

147. Q: Why MOSFET and IGBT are preferred switches in modern converters?

A: They are fully controlled and easy to drive.[1]

148. Q: What happens if a semi‑ controlled device is used where turn‑ off control is needed?

A: Large and complex commutation circuits are required.[1]

149. Q: Why BJTs lost popularity in high‑ power switching?

A: Need continuous base current and have slow turn‑ off.[1]

150. Q: What happens if we replace a BJT with MOSFET in a motor drive without redesigning gate/base drive?

A: Drive circuitry may be incompatible and MOSFET may not fully enhance.[1]

151. Q: Why IGBT dominates EV traction and industrial drives?

A: It balances high voltage capability with low conduction loss and reasonable switching speed.[1]

152. Q: Why MOSFET dominates low- voltage SMPS?

A: Extremely low RDS(on) and very fast switching at low voltages.[1]

"What if we swap device types?"

153. Q: What happens if we use a rectifier diode instead of fast recovery diode in SMPS?

A: Excess losses and possible transformer or device overheating.[1]

154. Q: What happens if we use MOSFET where SCR is normally used for controlled rectifier?

A: Control is possible but gate drive and snubber network become more complex.[1]

155. Q: What if an SCR is used instead of MOSFET in high- frequency chopper?

A: Switching frequency must be reduced drastically.[1]

156. Q: Why not use TRIAC instead of two MOSFETs in full- bridge inverter?

A: TRIAC cannot be turned OFF easily at arbitrary times; needs current zero.[1]

157. Q: What happens if we use BJT in place of IGBT in high- power inverter?

A: Base drive losses and switching losses become large.[1]

158. Q: What if we replace IGBT with MOSFET at 600 V and low current?

A: MOSFET may have higher RDS(on) but faster switching; choice depends on loss trade- off.[1]

159. Q: Why not use MOSFET for very high voltage (>3 kV) applications?

A: RDS(on) becomes too high due to thick drift region.[1]

160. Q: What happens if we use only diodes (no controlled devices) in converters?

A: No controllability over output except via passive means.[1]

Input polarity & misconnection questions

161. Q: What happens if we reverse polarity of supply to a bridge rectifier input?

A: DC output polarity remains same; bridge is symmetric.[1]

162. Q: What happens if AC supply is connected to DC side of bridge?

A: Diodes may be overstressed; circuit operation is undefined.[1]

163. Q: Why is gate always referenced to source/emitter?

A: Device behavior depends on gate‑source or gate‑emitter voltage, not absolute level.[1]

164. Q: What happens if gate‑source polarity is reversed on N‑channel MOSFET?

A: Negative VGS drives device OFF more strongly.[1]

165. Q: Why reverse polarity across body diode must be limited?

A: Exceeding its reverse rating causes breakdown.[1]

166. Q: What if load is connected to source instead of drain in low‑side MOSFET switch?

A: Still works but body diode orientation changes and performance may differ.[1]

167. Q: What happens if we connect MOSFET source to positive rail and drain to load to negative rail (P‑side connection for N‑channel)?

A: Body diode blocks correct conduction; cannot be driven as desired.[1]

168. Q: Why high‑side N‑MOSFET needs gate drive higher than supply?

A: To maintain VGS above threshold when source rides near supply.[1]

169. Q: What happens if high‑side gate is not driven high enough?

A: MOSFET operates in linear region, causing heating.[1]

170. Q: Why TRIAC gate polarity is flexible?

A: It can be triggered using various combinations of polarities relative to main terminals.[1]

Protection, limitations, and "what if we ignore…"

171. Q: What happens if we ignore maximum junction temperature?

A: Device lifetime shortens and failure risk rises sharply.[1]

172. Q: Why snubbers are used across power devices?

A: To limit dv/dt and overvoltage during switching.[1]

173. Q: What happens if dv/dt across SCR is too high?

A: It may turn ON unintentionally.[1]

174. Q: Why free‑wheel diode is used across inductive load with switch?

A: To provide path for inductive current and clamp voltage spikes.[1]

175. Q: What happens if free‑wheel diode is omitted?

A: Large voltage spikes can destroy the switch.[1]

176. Q: Why gate of MOSFET/IGBT must not float during faults?

A: Floating gate can allow unintended conduction.[1]

177. Q: What happens if reverse‑recovery current of diode is large?

A: It causes extra loss and stress in the main switch.[1]

178. Q: Why thermal runaway is a concern for BJTs but less for MOSFETs?

A: BJT current increases with temperature, while MOSFET RDS(on) tends to increase.[1]

179. Q: What happens if IGBT is heavily short‑circuited without protection?

A: High short‑circuit current causes rapid thermal destruction.[1]

180. Q: Why safe operating area (SOA) is specified?

A: To define voltage‑current limits where device can operate safely.[1]

Conceptual and interviewer‑style questions

181. Q: Why is a power switch ideally only ON or OFF, not linear?

A: To minimize power loss and heating.[1]

182. Q: What happens if a switch spends long time in transition region?

A: Switching losses increase significantly.[1]

183. Q: Why majority carriers give higher speed?

A: They do not need recombination to stop conduction.[1]

184. Q: What happens if both majority and minority carriers dominate conduction?

A: Device behaves slower but can have low on‑state voltage.[1]

185. Q: Why combining MOSFET gate with BJT conduction leads to IGBT?

A: It merges easy voltage drive with low on‑state loss at high voltage.[1]

186. Q: What happens to conduction losses if drift region doping is increased?

A: Resistance decreases, so conduction loss reduces.[1]

187. Q: What happens to blocking voltage if drift region doping is increased?

A: Breakdown voltage decreases.[1]

188. Q: Why there is a trade‑off between on‑state loss and blocking capability?

A: Because drift region thickness and doping affect both resistance and breakdown.[1]

189. Q: What happens if we try to make device both very low loss and very high voltage simultaneously?

A: Physical limits force large device size or complex structures.[1]

190. Q: Why power devices use vertical structures?

A: To conduct current vertically through thick drift layers for high current and voltage.[1]

191. Q: What happens if we only use planar (lateral) structure for high power?

A: Chip area becomes huge and conduction path is limited.[1]

192. Q: Why power MOSFET includes many cells in parallel internally?

A: To reduce RDS(on) and share current.[1]

193. Q: What happens if current crowding occurs in only few cells?

A: Local hotspots and premature failure.[1]

194. Q: Why negative temperature coefficient of MOSFET RDS(on) helps paralleling?

A: Hotter cell's resistance rises, pushing current to cooler cells.[1]

195. Q: What happens if BJTs are paralleled without emitter resistors?

A: One transistor may hog current and fail.[1]

196. Q: Why IGBTs also need careful paralleling?

A: Differences in VCEsat and temperature behavior affect current sharing.[1]

197. Q: What happens if gate‑drive timing in a leg is overlapping for both devices?

A: Shoot‑through occurs, causing large short‑circuit current.[1]

198. Q: Why dead‑time is introduced in complementary switches?

A: To avoid both devices conducting simultaneously.[1]

199. Q: What happens if dead‑time is too large?

A: Conduction through body diodes increases, raising losses and distortion.[1]

200. Q: Why understanding depletion, doping, and carrier types is essential for switch choice?

A: They determine speed, losses, and voltage capability.[1]

201. Q: As an interviewer, why ask "what if we change polarity or material" questions?

A: To check deep physical understanding, not just memorized formulas.[1]